In [1]:
```python
#IMPORTING LIBRARIES
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [4]:
```python
dataframe = pd.read_csv('Zomato data .csv')
```

In [5]:
```python
dataframe.head()
```

Out[5]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1/5	775	800	Buffet
1	Spice Elephant	Yes	No	4.1/5	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8/5	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	Buffet
4	Grand Village	No	No	3.8/5	166	600	Buffet

In [6]:
```python
dataframe.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype 
---  ------                       --------------  ----- 
 0   name                         148 non-null    object
 1   online_order                 148 non-null    object
 2   book_table                   148 non-null    object
 3   rate                         148 non-null    object
 4   votes                        148 non-null    int64 
 5   approx_cost(for two people)  148 non-null    int64 
 6   listed_in(type)              148 non-null    object
dtypes: int64(2), object(5)
memory usage: 8.2+ KB
```

CHANGING THE DATATYPE OF RATE COLUMN

In [7]:
```python
def handle_rate(value):
    value = str(value).split('/')
    value = value[0];
    return float(value)
dataframe['rate'] = dataframe['rate'].apply(handle_rate)
```

In [8]:
```python
# CHECKING FOR THE CHANGES
dataframe.head()
```

Out[8]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

In [9]:
```python
dataframe.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype  
---  ------                       --------------  -----  
 0   name                         148 non-null    object 
 1   online_order                 148 non-null    object 
 2   book_table                   148 non-null    object 
 3   rate                         148 non-null    float64
 4   votes                        148 non-null    int64  
 5   approx_cost(for two people)  148 non-null    int64  
 6   listed_in(type)              148 non-null    object 
dtypes: float64(1), int64(2), object(4)
memory usage: 8.2+ KB
```

In [12]:
```python
# checking for null values
dataframe.isnull().sum()
```

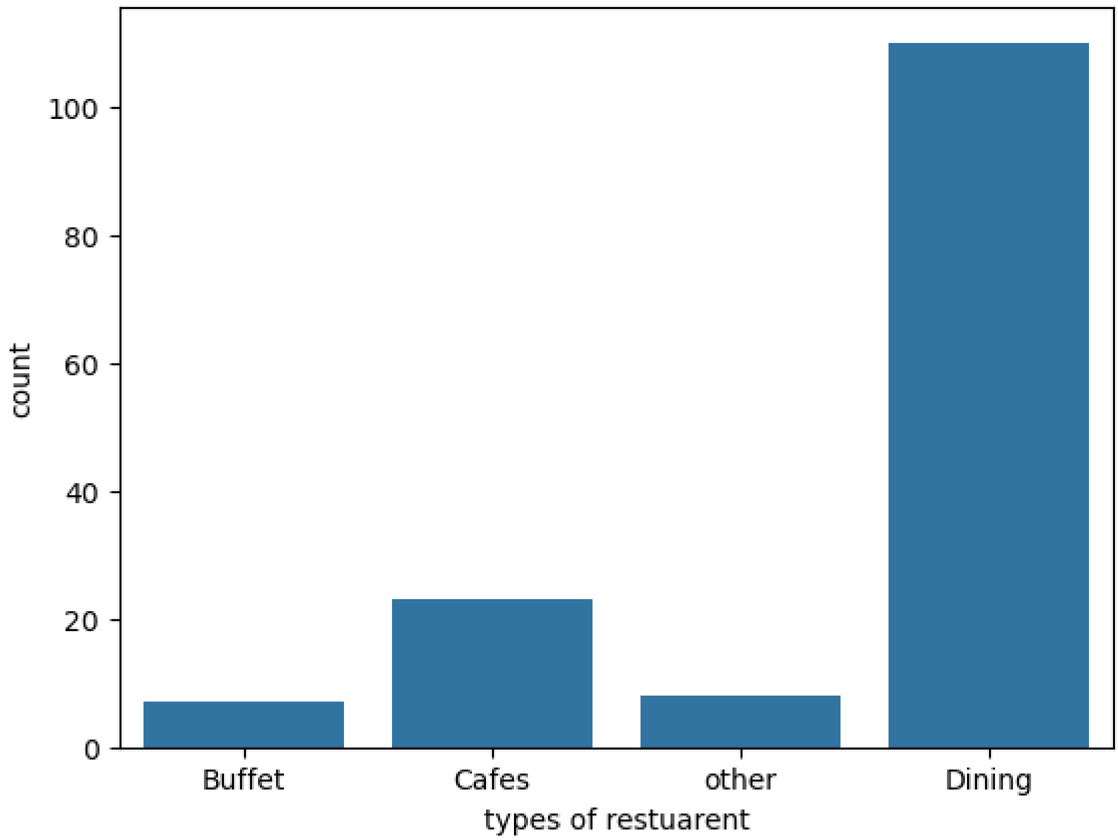
Out[12]:
```
name                           0
online_order                   0
book_table                     0
rate                           0
votes                          0
approx_cost(for two people)    0
listed_in(type)                0
dtype: int64
```

In [22]:
```python
dataframe['listed_in(type)'].sample(6)
```

Out[22]:
```
64     Dining
74     Dining
129    Dining
25      Cafes
88     Dining
20      Cafes
Name: listed_in(type), dtype: object
```

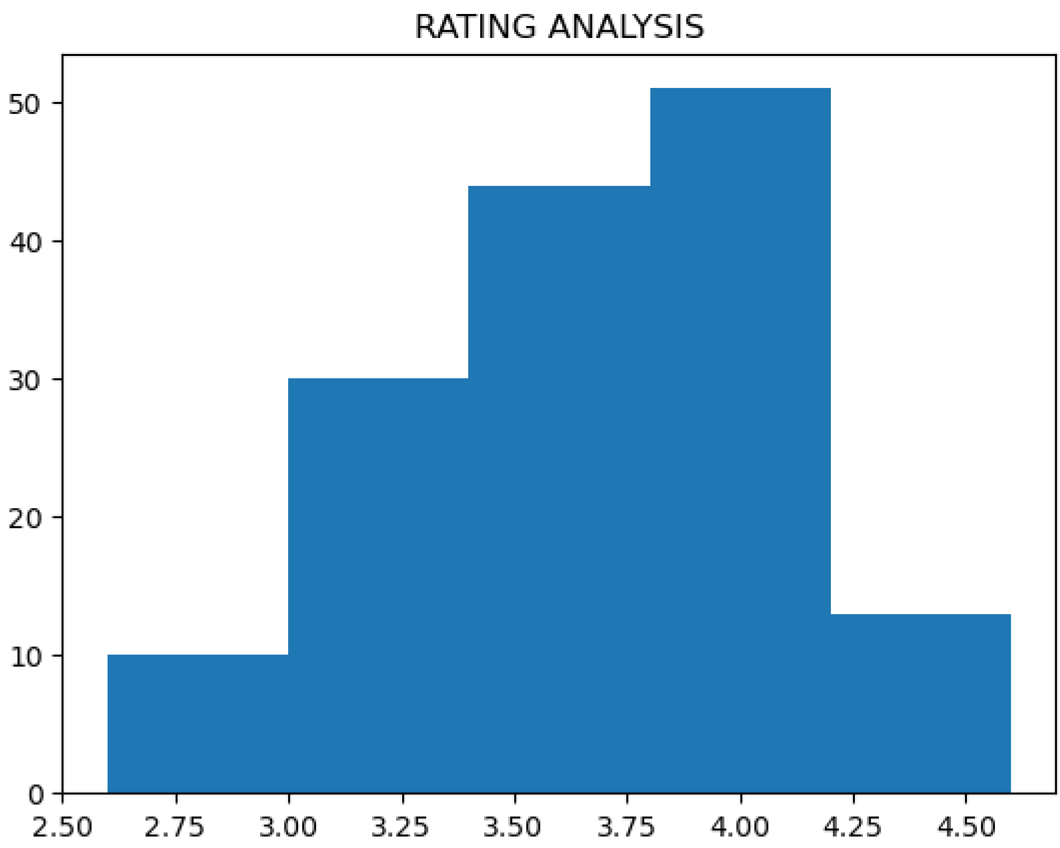
Q1. TYPES OF RESTUARENT

In [26]:
```
sns.countplot(x=dataframe['listed_in(type)'])
plt.xlabel('types of restuarent')
plt.show()
```



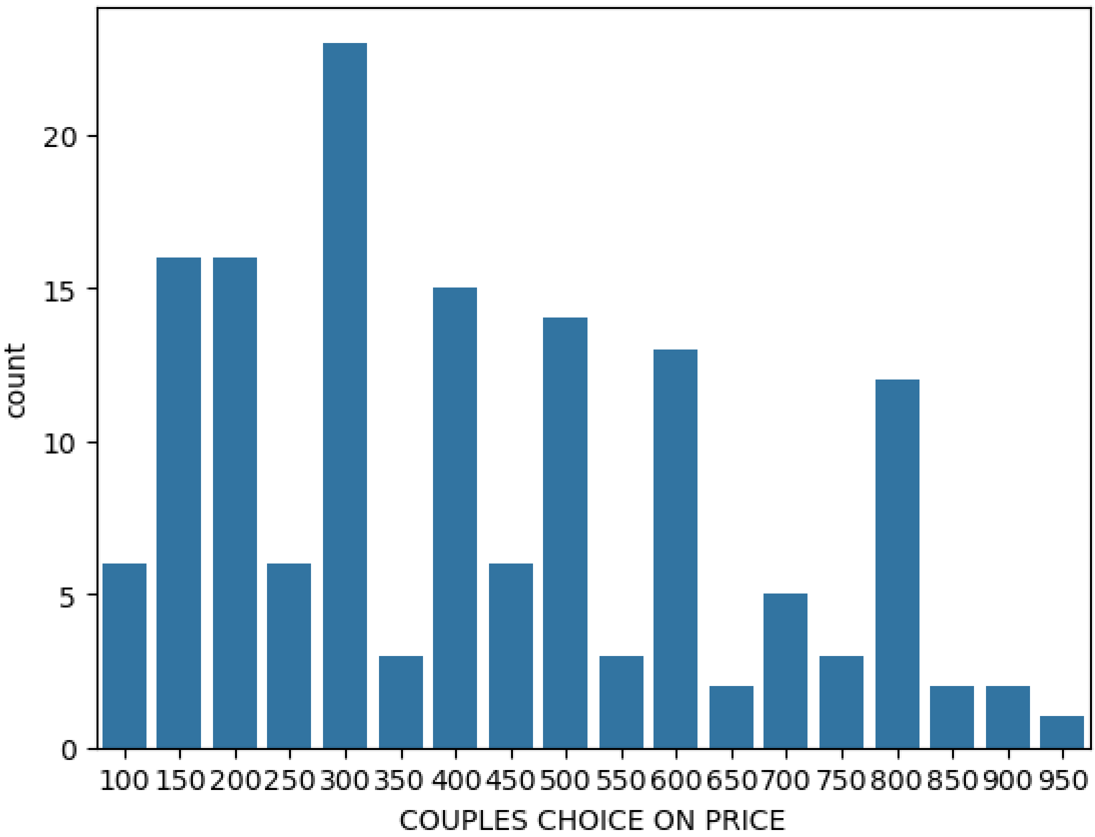
Q2. what are the majority ratings restuarant receive?

In [31]:
```
plt.hist(dataframe['rate'],bins = 5)
plt.title('RATING ANALYSIS')
plt.show()
```



WHAT IS THE APPROXIMATE PRICE COUPLES PREFER TO CHOOSE RESTUARANT

In [34]:
```
cost = dataframe['approx_cost(for two people)']
sns.countplot(x = cost)
plt.xlabel('COUPLES CHOICE ON PRICE')
plt.show()
```

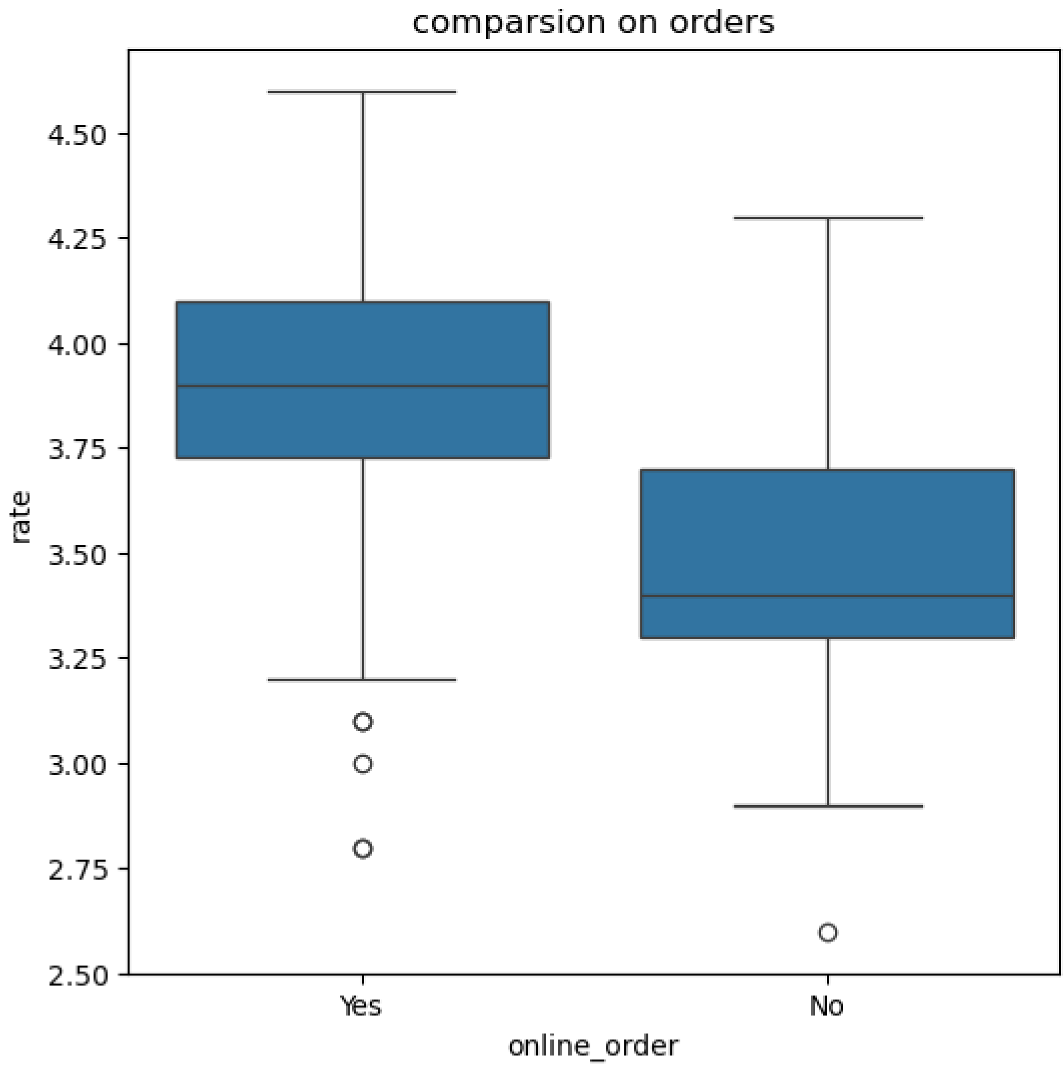


In [35]:
```
dataframe['online_order'].sample(3)
```

Out[35]:
```
69      Yes
138      No
110      No
Name: online_order, dtype: object
```

WHAT TYPE OF ORDERS RECEIVE MORE RATINGS?

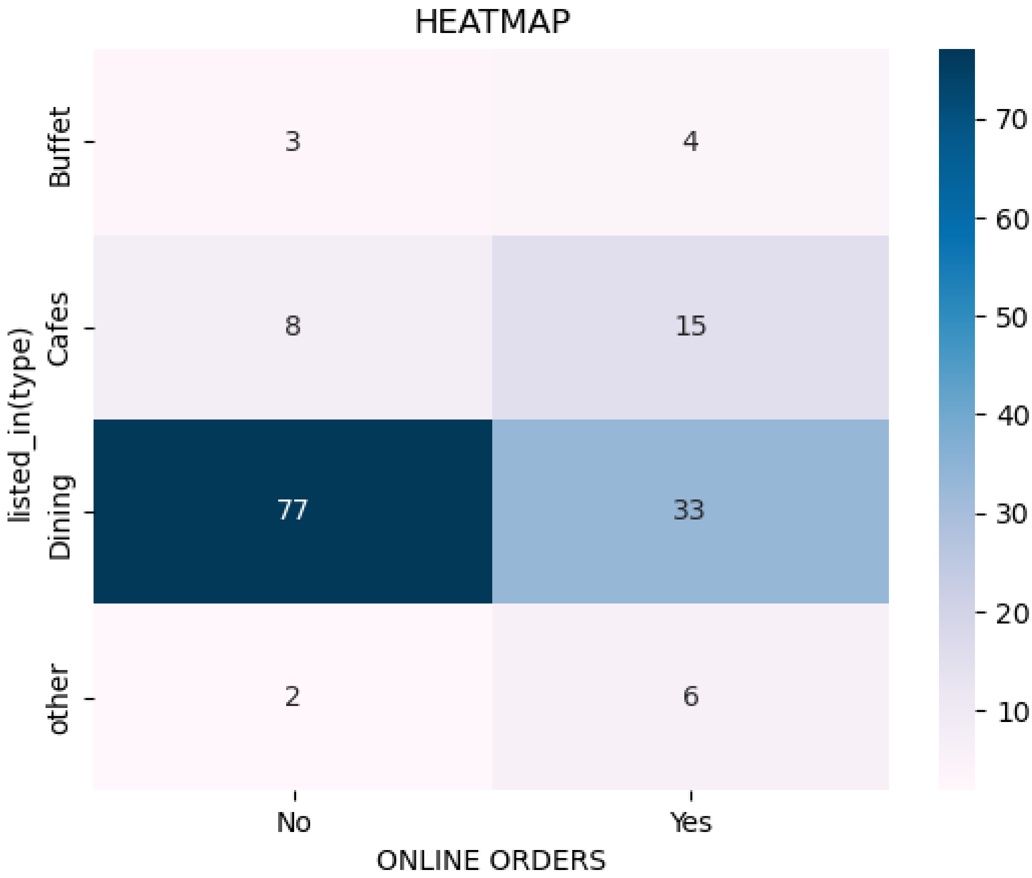
In [44]:
```
plt.figure(figsize=(6,6))
sns.boxplot(x ='online_order', y = 'rate',data = dataframe)
plt.title('comparsion on orders')
plt.show()
```



WHAT TYPE OF RESTUARANT RECEIVE ONLINE ORDERS,SO THAT ZOMATO CAN PROVIDE THOSE COSTUMERS WITH SOME GOD OFFERS?

In [50]:

```
pivot_table = dataframe.pivot_table(index = 'listed_in(type)', columns='o
sns.heatmap(pivot_table,annot = True, cmap ='PuBu',fmt ='d')
plt.title('HEATMAP')
plt.xlabel('ONLINE ORDERS')
plt.ylabel('listed_in(type)')
plt.show()
```


HEATMAP
Buffet
Cafes
Dining
other
listed_in(type)
3
4
8
15
77
33
2
6
No
Yes
ONLINE ORDERS
70
60
50
40
30
20
10